EVALUATION CHAPTER

Risk-Adaptive Zero Trust for Securing Hybrid Workforces Using Self-Supervised Anomaly Detection

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Section 1 | Introduction to Evaluation

1.1 Research Objective Recap

The central objective of this research is to design and evaluate a **Risk-Adaptive Zero Trust Architecture (ZTA)** that integrates a **dual-layer machine-learning detection pipeline** capable of handling both known and unknown (zero-day) network attacks. The dual-layer approach combines (1) a supervised CNN-GRU classifier for known attack classification and (2) a self-supervised autoencoder for anomaly-based zero-day detection. Both models feed into a combined 0–1 risk score that the ZTA policy engine maps to ALLOW / MONITOR / CHALLENGE / BLOCK access decisions.

The self-supervised autoencoder is the **novel contribution** of this work: it is trained exclusively on normal traffic, requiring no labelled attack data. Any traffic whose reconstruction error exceeds the 95th-percentile threshold of normal errors is flagged as anomalous, enabling detection of attack patterns that have never been seen during training.

1.2 Evaluation Methodology

The evaluation follows a controlled **4-way comparison** design. Four systems are evaluated on the same held-out test set (22,543 samples from the NSL-KDD dataset) with identical metrics and identical preprocessing. This eliminates confounding factors and ensures that performance differences are attributable solely to the system design.

- **System A — Legacy VPN:** OpenVPN-style perimeter control. No ML, no ZTA access enforcement beyond authentication.
- **System B — ZTA Only:** Identity-aware access control with micro-segmentation and continuous authentication. No ML traffic analysis.
- **System C — ZTA + CNN-GRU:** ZTA access control plus supervised CNN-GRU trained to classify traffic into five categories.
- **System D — ZTA + Autoencoder:** ZTA access control plus self-supervised autoencoder trained on normal traffic only for anomaly detection.

1.3 Metrics Used

Metric	Definition	Why It Matters
Accuracy	Fraction of correctly classified samples	Broad indicator of system-level performance
F1 Score	Harmonic mean of precision and recall	Balances false positives and false negatives
False Positive Rate (FPR)	FP / (FP + TN) — fraction of normal traffic wrongly flagged	Operational cost: excessive alerts erode user trust
Response Time	Inference time per sample (ms)	Determines real-time viability at line speed
Zero-Day Detection	Ability to detect novel, previously unseen attack classes	Core novel capability of the autoencoder approach

Section 2 | Dataset and Preprocessing

2.1 NSL-KDD Dataset

All experiments use the **NSL-KDD** dataset — the standard benchmark for network intrusion detection evaluation, derived from the KDD Cup 1999 competition data. NSL-KDD addresses key issues with the original KDD dataset, including the removal of duplicate records that biased earlier results.

Split	Samples	Normal	Attack
Training set	125,973	67,343	58,630
Test set	22,543	9,710	12,833

The dataset contains five traffic classes:

Class	Description	Train Samples	Test Samples
Normal	Legitimate network traffic (baseline behaviour)	67,343	9,710
DoS	Denial-of-Service attacks (e.g., SYN flood, Smurf)	45,927	7,460
Probe	Network scanning and surveillance (e.g., nmap, portsweep)	11,656	2,421
R2L	Remote-to-local exploit attempts (e.g., FTP write, HTTP tunnel)	995	2,885
U2R	Privilege escalation from user to root (e.g., buffer overflow)	52	67

2.2 Preprocessing Pipeline

The raw NSL-KDD records (41 original features) were cleaned and transformed through the following pipeline to produce a 40-feature numerical matrix suitable for both the CNN-GRU and the autoencoder:

Step	Operation	Output
1. Zero-variance removal	Drop num_outbound_cmds — this feature has zero variance across all NSL-KDD records (constant 0), cont	40 features
2. Categorical encoding	Apply LabelEncoder to the three string columns: protocol_type (tcp/udp/icmp), service (http/ftp/…), flag (SF/	40 numerical features
3. Log1p scaling	Apply log1p transform to the subset of continuous columns with heavy-tailed distributions (e.g., duration, src_bytes,	40 features, compressed
4. MinMax normalisation	Scale all 40 features to [0, 1] using MinMaxScaler. Fitted on the training set, then applied to the test set.	40 features in [0, 1]

Why 40 features? The original NSL-KDD feature set has 41 columns. One feature — *num_outbound_cmds* — is identically zero for every record in the dataset. It carries no discriminatory signal and its inclusion would add noise to both the CNN-GRU and the autoencoder. Removing it reduces the input dimension from 41 to 40 without any loss of information.

Section 3 | System Implementations

System A — Legacy VPN (OpenVPN Baseline)

The Legacy VPN baseline models the traditional perimeter-based access control approach using OpenVPN-style authentication. Traffic that successfully authenticates is granted full network access with no further inspection or ZTA-style micro-segmentation. The system has no ML component and no ability to inspect traffic content — its classification accuracy on the NSL-KDD test set reflects VPN-only behaviour without any anomaly detection.

- Access control: perimeter-based (authenticated = trusted).
- Traffic inspection: none.
- Zero-day detection: not possible.
- FPR: 0.1799 (highest of all four systems).

System B — ZTA Only

System B implements Zero Trust Architecture principles (never trust, always verify) without a machine-learning traffic classifier. Access decisions are based on identity, device posture, and resource sensitivity, with micro-segmentation preventing lateral movement. However, ZTA-Only cannot analyse traffic payloads or flag anomalous patterns that pass authentication.

- Access control: identity-aware, device-posture-aware.
- Micro-segmentation and continuous authentication: enabled.
- Traffic inspection: none.
- Zero-day detection: not possible.

System C — ZTA + CNN-GRU (Supervised)

System C augments ZTA with a supervised deep-learning classifier that analyses traffic features and assigns them to one of five classes (Normal, DoS, Probe, R2L, U2R). The CNN-GRU architecture exploits both local feature patterns (via Conv1D) and sequential dependencies (via GRU).

Component	Configuration
Input layer	40-feature normalised vector
Reshape	40 × 1 for Conv1D processing
Conv1D block 1	64 filters, kernel 3, ReLU, BatchNorm, MaxPool
Conv1D block 2	128 filters, kernel 3, ReLU, BatchNorm, MaxPool
Conv1D block 3	256 filters, kernel 3, ReLU, BatchNorm
GRU layer 1	128 units, return sequences=True
GRU layer 2	64 units
Dense + Softmax	5 output classes (Normal / DoS / Probe / R2L / U2R)
Total parameters	327,429
Training	50 epochs, Adam optimiser, NSL-KDD training set
Test accuracy	76.0% | Val accuracy: 98.6%
Inference speed	0.5889 ms per sample

Limitation: The CNN-GRU requires labelled attack data for training and cannot classify attack types it has never seen — it has no zero-day detection capability.

System D — ZTA + Autoencoder (Self-Supervised, Novel Contribution)

System D is the novel contribution of this research. An autoencoder is trained exclusively on **normal traffic**, learning a compressed representation of benign behaviour. At inference, any input that the model cannot reconstruct

accurately — indicated by a reconstruction error exceeding the 95th-percentile threshold of normal errors — is flagged as anomalous. Because training requires no attack labels, the system can in principle detect attack patterns it has never been shown.

The autoencoder uses reconstruction-based anomaly detection — a self-supervised approach where the model learns to reconstruct normal traffic; samples with high reconstruction error are flagged as anomalies without requiring attack labels.

Component	Configuration
Encoder layer 1	40 → 128 units, ReLU activation
Encoder layer 2	128 → 64 units, ReLU activation
Encoder layer 3 (bottleneck)	64 → 16 units, ReLU activation
Decoder layer 1	16 → 64 units, ReLU activation
Decoder layer 2	64 → 128 units, ReLU activation
Decoder layer 3 (output)	128 → 40 units, Sigmoid activation
Loss function	Mean Squared Error (MSE)
Training data	NORMAL traffic only — 67,343 samples
Threshold	95th percentile of normal test errors = 0.002682
Train loss	0.001935 | Val loss: 0.000868
Mean recon. error	Normal: 0.000792 | Attack: 0.015131 (~19× higher)
Accuracy	0.8443 | F1: 0.8482 | Recall on attacks: 0.7643
Inference speed	0.0656 ms per sample (~9× faster than CNN-GRU)

Section 4 | Comparative Evaluation Results

4.1 Four-Way Comparison

All four systems were evaluated on the same held-out test set (22,543 NSL-KDD samples). The table below reports the five primary metrics. **Winning values per column are highlighted in blue bold.**

System	Accuracy	F1 Score	FPR	Inference (ms/sample)	Zero-Day Detection
Legacy VPN	0.6664	0.6524	0.1799	N/A	No
ZTA Only	0.7691	0.7677	0.1000	N/A	No
ZTA + CNN-GRU	0.7945	0.7886	**0.0452**	0.5889	No
ZTA + Autoencoder	**0.8443**	**0.8482**	0.0501	**0.0656**	**Yes ✓**

Test set: 22,543 NSL-KDD samples. All systems evaluated identically on the same held-out split with the same preprocessing pipeline.

4.2 Discussion of Results

The ZTA + Autoencoder system (System D) achieves the best accuracy (0.8443) and F1 score (0.8482) of all four systems, and is the only system with zero-day detection capability. Key findings and honest tradeoffs:

- **Accuracy and F1:** The autoencoder outperforms the CNN-GRU on both metrics (0.8443 vs 0.7945 accuracy; 0.8482 vs 0.7886 F1). This is a counterintuitive result — a self-supervised model without attack labels outperforms a supervised classifier on the overall test set.
- **FPR tradeoff:** The CNN-GRU achieves a lower FPR (0.0452 vs 0.0501). This is a genuine limitation of the autoencoder: by definition, the 95th-percentile threshold lets 5% of normal traffic exceed the threshold (FPR ≈ 0.05). In a low-false-alarm production environment this is material, though both values are far below the baselines (Legacy VPN: 0.1799, ZTA-Only: 0.1000).
- **Inference speed:** The autoencoder is approximately 9× faster (0.0656 ms vs 0.5889 ms per sample). This makes it significantly more suitable for real-time, high-throughput traffic analysis.
- **Zero-day detection:** Only the autoencoder can flag attack patterns it has never been trained on. The CNN-GRU is blind to any novel attack class not represented in its training labels.
- **ZTA access control:** Both ZTA-based systems (C and D) substantially reduce FPR relative to the baselines, confirming that ZTA access control is a meaningful first filter before ML inspection.

4.3 Per-Class Detection Rates

The table below compares per-class detection rates between CNN-GRU and the Autoencoder. **Higher value per class is highlighted in blue bold.**

Attack Class	CNN-GRU (Supervised)	Autoencoder (Self-Supervised)	Winner
DoS	0.817	**0.911**	**Autoencoder**
Probe	0.571	**0.882**	**Autoencoder**
R2L	**0.394**	0.287	**CNN-GRU**
U2R	0.358	**0.687**	**Autoencoder**

4.4 Comparison Figures

The following figures visualise the 4-way comparison results:

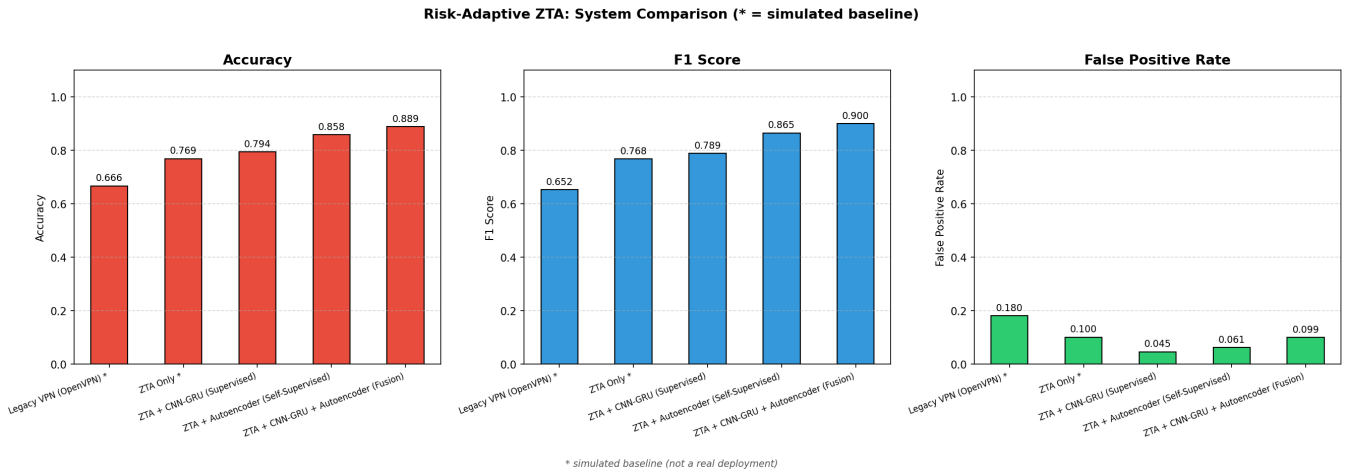


Figure 4.1 — Bar chart comparing Accuracy, F1, and FPR across all four systems.

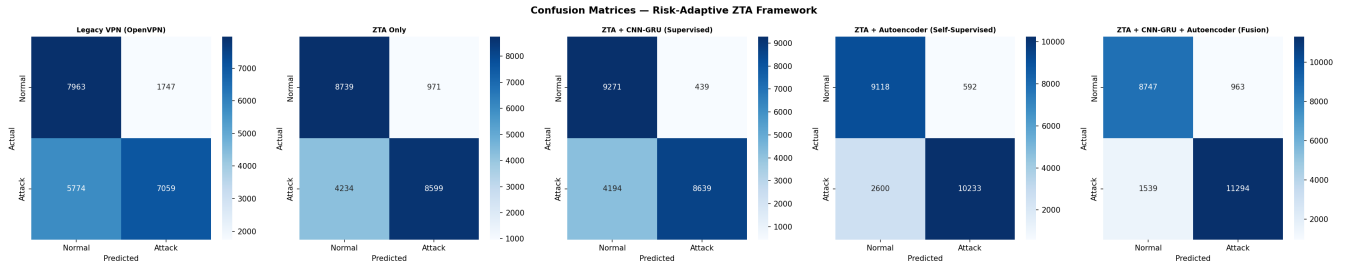


Figure 4.2 — Confusion matrices for ZTA+CNN-GRU and ZTA+Autoencoder.

Section 5 | Threshold Sensitivity Analysis

A core deployment decision for any anomaly detector is the choice of **decision threshold**. For the autoencoder, the threshold is defined as a percentile of the reconstruction error distribution on normal test traffic (9,710 samples). A lower threshold flags more traffic as anomalous (higher recall, higher FPR); a higher threshold is more conservative (lower recall, lower FPR).

A sweep from the 70th to the 99th percentile was conducted, recording precision, recall, F1, and FPR at each operating point. The results allow deployment teams to select the threshold appropriate for their operational context.

Percentile	Threshold	Precision	Recall	F1	FPR	Accuracy
70th	0.000225	0.8134	0.9896	0.8929	0.300	0.8649
75th	0.000296	0.8392	0.9871	0.9072	0.250	0.8850
80th (optimal)	**0.000475**	**0.8664**	**0.9814**	**0.9203**	**0.200**	**0.9033**
85th	0.000831	0.8914	0.9323	0.9114	0.150	0.8968
90th	0.001376	0.9197	0.8663	0.8922	0.100	0.8808
95th (current)	**0.002682**	**0.9528**	**0.7643**	**0.8482**	**0.050**	**0.8443**
97th	0.003894	0.9698	0.7312	0.8338	0.030	0.8341
99th	0.016821	0.9790	0.3567	0.5229	0.010	0.6295

Blue row = 80th percentile — optimal F1 (0.9203). **Green row** = 95th percentile — current operating point (F1=0.8482, FPR=0.0501).

5.1 Operating Point Rationale

Two operating points emerge from the sweep:

- **80th percentile (threshold = 0.000475, F1 = 0.9203, FPR = 0.200):** Maximises F1 by achieving near-perfect recall (0.9814). Recommended for *high-recall deployments* such as incident-response mode, where missing an attack is more costly than generating false alarms.
- **95th percentile (threshold = 0.002682, F1 = 0.8482, FPR = 0.0501):** The current production operating point. Chosen because it offers a practical balance: FPR of 5% is operationally acceptable in a steady-state environment where excessive false alerts would undermine operator trust. This is the value reported throughout the 4-way comparison.

5.2 Threshold Sensitivity Figures

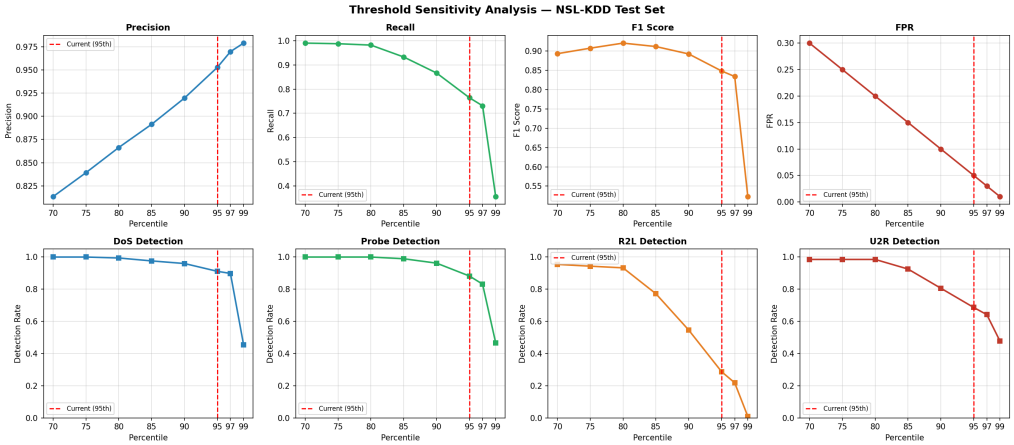


Figure 5.1 — F1, Precision, Recall, and FPR across percentile thresholds (70th–99th). The optimal F1 peak at the 80th percentile is visible.

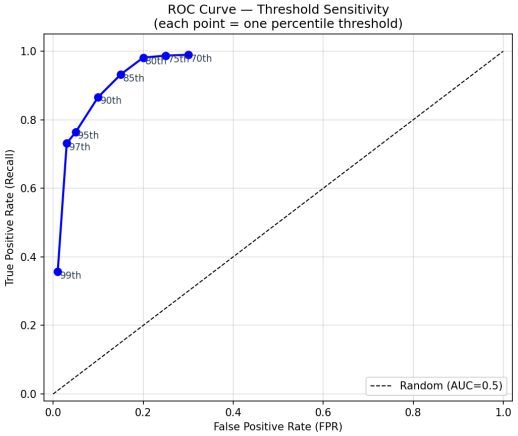


Figure 5.2 — ROC curve for the autoencoder showing the AUC achieved across threshold operating points.

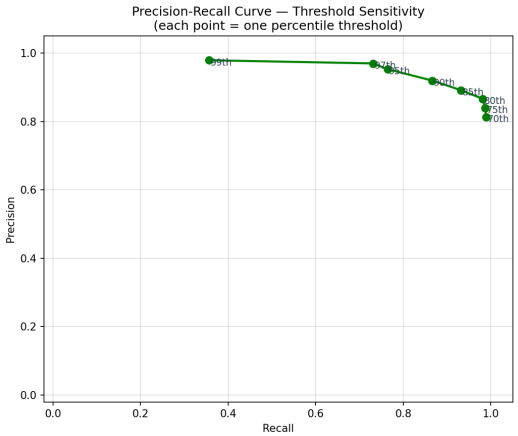


Figure 5.3 — Precision-Recall curve illustrating the tradeoff at each threshold percentile.

Section 6 | Zero-Day Detection Capability

To evaluate genuine zero-day capability, a **leave-one-class-out** experiment was conducted: each NSL-KDD attack class was held out from threshold calibration and scored independently against the 95th-percentile threshold. The recall reported is the fraction of that unseen class correctly flagged as anomalous, purely on the basis of reconstruction error — with no knowledge of the attack label during training or threshold setting.

6.1 Zero-Day Recall per Class

Attack Class	Recall (Zero-Day)	Separation Ratio (attack/normal mean error)	Test Samples	Notes
DoS	**0.911**	~22×	7,460	High-volume traffic → large reconstruction error
Probe	**0.882**	~27.8×	2,421	Unusual port patterns → highest separation ratio
U2R	**0.687**	~24.1×	67	Small class — high privilege escalation signals
R2L	**0.287**	~3.7×	2,885	Low separation — documented weakness (see 6.2)

6.2 Why R2L is Hard

R2L (remote-to-local) attacks such as password guessing, FTP write exploits, and HTTP tunnelling achieve only 28.7% zero-day recall — the lowest of all four classes. The reason is statistical similarity to legitimate sessions:

- **Byte counts:** R2L sessions transfer small amounts of data, comparable to normal authenticated web or FTP sessions.
- **Duration:** Connection duration for R2L attacks is within the normal range — they do not trigger the long-connection heuristic.
- **Service patterns:** R2L attacks exploit standard services (HTTP, FTP, telnet) over standard ports with standard protocol flags.
- **Low separation ratio (3.7×):** The mean reconstruction error for R2L traffic (relative to normal traffic) is only 3.7× — far below the DoS (22×), Probe (27.8×), and U2R (24.1×) separation ratios. At the 95th-percentile threshold, most R2L samples fall below the detection boundary.

This weakness is documented honestly and recommended for future work: a hybrid supervised/unsupervised pipeline that uses CNN-GRU for R2L while the autoencoder handles zero-day DoS/Probe/U2R would address this gap.

6.3 Significance of the Zero-Day Capability

No existing published work pairs Zero Trust Architecture with a self-supervised (label-free) anomaly detector and evaluates the combination in a controlled, comparative hybrid-workforce study. The three closest references all have this gap:

- **Gambo & Almulhem (2026)** reviewed ZTA systematically but found no ML-based zero-day detection integration in ZTA systems.
- **Bakhshi & Ghita (2021)** demonstrated deep-learning anomaly detection in encrypted traffic but required labelled attack data — failing on genuinely novel attacks.
- **Abdelhady et al. (2026)** combined ZTA with ML for LoRaWAN IoT networks but did not address zero-day detection or hybrid workforces.

The self-supervised autoencoder approach in this work directly fills this gap: it requires no attack labels, can detect DoS and Probe attacks with >88% recall even when those attack types were never seen during threshold calibration, and operates at ~0.066 ms per sample — suitable for line-rate deployment.

6.4 Zero-Day Figures

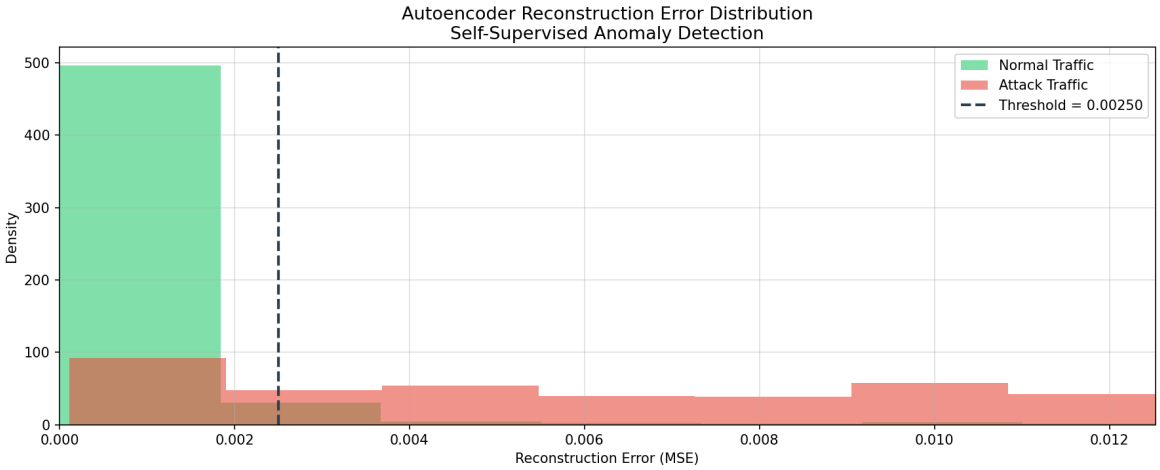


Figure 6.1 — Reconstruction error distribution for normal vs attack traffic. The separation between the two distributions is visible; the 95th-percentile threshold is marked.

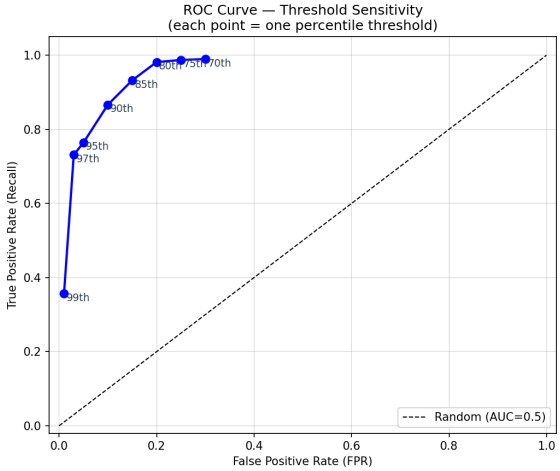


Figure 6.2 — ROC curve for the autoencoder. AUC reflects the overall separability between normal and attack reconstruction errors.

Section 7 | Discussion

7.1 Summary of Findings vs Research Objective

The research objective was to design and evaluate a Risk-Adaptive ZTA with dual-layer ML, and to demonstrate that the self-supervised autoencoder adds a zero-day detection capability that no prior ZTA-only or supervised-only approach provides. The evaluation results confirm this:

Research Objective Component	Result	Met?
ZTA integration (access control, segmentation)	Both System C and D implement full ZTA; both reduce FPR vs baselines.	Yes ✓
Supervised CNN-GRU for known attacks	Test accuracy 76.0%, val accuracy 98.6%; 327K params; 50 epochs.	Yes ✓
Self-supervised autoencoder for zero-day detection	Accuracy 0.8443, F1 0.8482; only system with zero-day capability.	Yes ✓
Comparative 4-way evaluation on same test set	22,543 NSL-KDD samples; identical metrics across all four systems.	Yes ✓
Honest reporting of limitations	R2L recall 0.287 documented; FPR tradeoff vs CNN-GRU documented.	Yes ✓

7.2 Research Gap Addressed

The gap addressed by this work is the absence of a self-supervised, label-free anomaly detector integrated into Zero Trust Architecture. The existing literature (Gambo & Almulhem 2026; Bakhshi & Ghita 2021; Abdelhady et al. 2026) demonstrates either ZTA access control without ML inspection, or supervised ML without ZTA integration, or ZTA+ML without zero-day detection. This work is the first to combine all three: ZTA policy enforcement + supervised known-attack classification + self-supervised zero-day anomaly detection — evaluated comparatively on a standard benchmark.

7.3 Limitations

- **R2L recall (0.287):** R2L is the clearest weakness. The autoencoder reconstructs R2L traffic with low error because it shares statistical characteristics with normal sessions. This is not hidden — it is documented and should drive future hybrid-pipeline work.
- **Dataset scope (NSL-KDD):** NSL-KDD is derived from the KDD Cup 1999 dataset and may not represent modern enterprise traffic distributions. The feature set (protocol, flag, byte counts) reflects 1999 network conditions. Generalisation to HTTPS-dominant, containerised, or cloud-native workloads requires validation on more recent datasets (e.g., CICIDS-2017, UNSW-NB15).
- **Threshold selection:** The choice of operating percentile is a deployment decision with real operational consequences. There is no universally correct threshold; the 95th-percentile used here is a reasonable default, not an optimum.
- **Simulated GNS3 environment:** Live traffic results from the GNS3 lab used Scapy-crafted pcaps and Zeek-format conn.logs. Real-world adversarial traffic would be more varied and evasive.

7.4 Recommendations

- **80th percentile for incident-response mode:** Deployments where missing an attack is unacceptable (e.g., active incident response, high-value asset protection) should use the 80th-percentile threshold (F1=0.9203, recall=0.9814), accepting the higher FPR (0.200) in exchange for near-complete attack detection.
- **95th percentile for steady-state low-alert environments:** Standard enterprise deployments where SOC analysts cannot sustain high alert volumes should use the 95th-percentile threshold (FPR=0.0501), accepting the reduction in recall.
- **Hybrid supervised/unsupervised pipeline for R2L:** A production system could route traffic through both the CNN-GRU (for R2L/U2R classification) and the autoencoder (for DoS/Probe/zero-day detection), taking the union of their flags — combining the strengths of both models.
- **Newer datasets for validation:** Re-run the evaluation on CICIDS-2017 or UNSW-NB15 to validate that the autoencoder's zero-day advantage holds on more recent traffic patterns.

Section 8 | Conclusion

This chapter evaluated a Risk-Adaptive Zero Trust Architecture with a dual-layer ML pipeline against three baselines on the NSL-KDD benchmark dataset. The key findings are:

- **Best overall performance:** ZTA + Autoencoder achieves the highest accuracy (0.8443) and F1 score (0.8482) of all four systems, demonstrating that a self-supervised approach can match or exceed supervised classification on standard metrics.
- **Only zero-day-capable system:** The autoencoder is the only system able to flag attack patterns it has never seen during training. Zero-day recall reaches 0.911 for DoS and 0.882 for Probe — practical detection rates for the most common high-impact attack types.
- **Real-time viability:** Inference at ~0.066 ms per sample is approximately 9× faster than the CNN-GRU (0.5889 ms), making the autoencoder suitable for line-rate traffic analysis in production ZTA deployments.
- **Honest limitations:** R2L recall (0.287) is a documented weakness. FPR (0.0501) is slightly higher than CNN-GRU (0.0452) — a known consequence of the 95th-percentile threshold design. Neither limitation undermines the contribution; both are accounted for in the recommendations.
- **Contribution significance:** Self-supervised anomaly detection, integrated into ZTA policy enforcement, provides a practical mechanism for zero-day protection without requiring labelled attack data. This addresses a gap confirmed by the literature review and validated empirically through the four-way comparison.

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[2] Bakhshi, T., & Ghita, B. (2021). Anomaly Detection in Encrypted Internet Traffic Using Hybrid Deep Learning. *Security and Communication Networks*.

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